

Evaluation Report

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Abstract. This document reports the evaluation metrics for the gommage explainability middleware. The evaluation compares agent execution before and after middleware integration and measures response latency, token overhead for reasoning provenance, and replay fidelity. The dataset was manually generated from 10 execution scenarios using `gpt-o4-mini` as the agent LLM with temperature set to 0.85.

Evaluation Setup

The evaluation was designed to measure whether adding the AER-generating middleware changes the practical behavior of the agent while still producing useful reasoning provenance.

- **Execution scenarios:** 10 manually generated agent execution scenarios.
- **LLM:** `gpt-o4-mini`.
- **Temperature:** 0.85.
- **Comparison:** before middleware integration versus after middleware integration.
- **Recorded artifact:** Agent Execution Record objects containing reasoning provenance.

Metrics

The evaluation focuses on three metrics:

1. **Response latency:** the average change in response latency after middleware integration compared with the baseline agent execution.
2. **Token overhead:** the average additional token cost required to generate AER reasoning provenance.
3. **Replay fidelity:** the percentage of runs whose replay preserved the recorded execution trajectory across the evaluated scenarios.

Results

Metric	Result	Interpretation
Response latency delta	6 ms average delta across 10 runs; p = 0.24	The measured latency difference before and after middleware integration was not statistically significant.
Token overhead	40 tokens on average	Generating AER objects introduced a small average overhead for reasoning provenance capture.
Replay fidelity	100% across 10 runs	All evaluated executions replayed successfully against their recorded trajectory.

Latency Interpretation

The average response latency delta was 6 ms across 10 runs. The associated p-value was 0.24, which does not provide evidence of a statistically significant latency difference between the baseline agent and the middleware integrated agent. In this evaluation, the middleware did not introduce a meaningful latency penalty.

Token Overhead Interpretation

The average token overhead for generating AER objects was 40 tokens. This overhead corresponds to the additional structured reasoning provenance emitted for each execution. The measured value indicates that the explainability layer adds compact provenance metadata rather than a large auxiliary trace.

Replay Fidelity Interpretation

Replay fidelity was 100% across the 10 manually generated scenarios. This means that each recorded execution could be replayed from its AER without divergence from the recorded trajectory under the evaluation procedure.

Conclusion

The evaluation shows that gomme provides reasoning provenance through AER generation while preserving practical execution performance. The latency delta was small and not statistically significant, the token overhead averaged 40 tokens, and replay fidelity reached 100% across the evaluated scenarios.